

LC 448: Find All Numbers Disappeared in an array

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Adobe, Oracle

Goal:

basically we need to return array of numbers which don't appear in given array of $[1, n]$

Optimal Approach

Optimal Approach without extra space.

Core Idea: Utilize the gaps.

Example Array:

[1 2 2 3 3 4 7 8]

Algorithm Steps

```
start from curr = 1
```

```
then run a loop from 0 to n
```

```
let the num be nums[i]
```

```
then diff = nums[i] - curr
```

```
if (<0) ignore
```

```
but if (>0) it means there are gaps
```

C++ Code Implementation

```
class Solution {
public:
    vector<int> findDisappearedNumbers(vector<int>& nums) {

        vector<int> ans;
        int n = nums.size();

        sort(nums.begin(), nums.end());
        int curr = 1;

        for(int i = 0; i < n; i++) {
            int num = nums[i];
            int diff = num - curr;

            if(diff < 0) continue;
            if(diff > 0) {
                while(diff-- > 0) {
                    ans.push_back(curr);
                    curr++;
                }
            }

            curr = num + 1;
        }

        // what if peeche ke udde hue ho

        while(curr <= n) {
            ans.push_back(curr);
            curr++;
        }

        return ans;
    }
};
```